

Diagram showing the UART4 pin connections to the ELRS module:

- UART4_TX_TSVR is connected to TSVR_RX.
- UART4_RX_TSVR is connected to TSVR_TX.
- The module is powered by +5V and GND.

File: elrs.kicad_sch

The diagram shows a component symbol for a barometer. It has three inputs on the left: `I2C1_SCL_Sense` (green), `I2C1_SDA_Sense` (green), and `X` (blue). The inputs are connected to pins `Baro_SCL`, `Baro_SDA`, and `Baro_DR1` respectively. The component has two outputs on the right: `+3.3V` (green) and `GND` (green), which are connected to external `+3.3V` and `GND` power pins. The component is labeled `Barometer` at the top and `File: barometer.kicad_sch` at the bottom.

Pin	Connection
<u>OpF_NCS</u>	>OpF_NCS +3.3V → +3V3
<u>SPI2_MISO_OpF</u>	>OpF_MISO GND → GND
<u>SPI2_MOSI_OpF</u>	>OpF_MOSI LDO_EN → OpF_LDO_EN
<u>SPI2_SCK_OpF</u>	>OpF_SCLK
<u>OpF_MOTION</u>	>OpF_MOTION
<u>OpF_NRESET</u>	>OpF_NRESET

File: opticalflow.kicad_sch

A schematic diagram of a 100nF capacitor array. It consists of a horizontal row of 11 capacitors, labeled C80 through C91. The top plate of each capacitor is connected to a +3V supply, indicated by a red arrow pointing to the top line. The bottom plate of each capacitor is connected to a common GND, indicated by a red arrow pointing to the bottom line. Each capacitor is represented by two parallel horizontal lines, with the value '100n' written below each pair. The capacitors are connected in series along the top and bottom lines.

IO: G6-G11 & G15

IO: F2 & F11-F15
Analog: F3-F6 & F10

IO: E3, E4, & E9-E15

IO: C15
Analog: PC0-4

IO: D7, & D10-D15

STM32F427ZGTX

12VBuckEN

Power

12VBuckEN

File: power.kicad_sch

ST-Conn

Pin	Signal
1	+3V3MCU
2	+3V3MCU
3	×
4	SWCLK
5	-3V3MCU
6	-3V3MCU
7	NRST
8	SWDIO

The diagram illustrates the wiring for two servos, G1 (Gimbal1) and G2 (Gimbal2). Each servo has three pins: 3 (red), 2 (green), and 1 (blue). The red pin (3) is connected to +5V, the green pin (2) is connected to GND, and the blue pin (1) is connected to GND.

File: tof.kicad_sch

Pin connection diagram for IMU1 and IMU2. The diagram shows the connections for two IMU modules (IMU1 and IMU2) to a microcontroller. The connections are as follows:

- IMU1 Connections:**
 - IMU1_CS to IMU_CS (+3.3V)
 - IMU1_INT1 to IMU_INT1 (GND)
 - IMU1_MISO to IMU_SDO
 - IMU1_MOSI to IMU_SDI
 - IMU1_SCK to IMU_SCLK
 - IMU1_FSYNC to IMU_FSYNC
- IMU2 Connections:**
 - IMU2_CS to IMU2_CS (+3.3V)
 - IMU2_INT1 to IMU_INT1 (GND)
 - IMU2_MISO to IMU_SDO
 - IMU2_MOSI to IMU_SDI
 - IMU2_SCK to IMU_SCLK
 - IMU2_FSYNC to IMU_FSYNC

File: imu.kicad_sch

Diagram showing the SDMicro component (red box) connected to the SD_CS, SPI3_MISO_SD, SPI3_MOSI_SD, and SPI3_SCK_SD pins. The component is labeled "SDMicro". The pins are connected to the SD_CS, SD_MISO, SD_MOSI, and SD_CLK pins of the component. The SD_MISO pin is connected to +3.3V, and the SD_MOSI pin is connected to GND. The SD_CLK pin is connected to GND. The file name "File: sdmicro.kicad_sch" is shown at the bottom.

J21 Supplies

Pin	Voltage
1	+3V3
2	+5V
3	GND
4	+5V
5	+3V3
6	+5V
7	+3V3
8	+5V
9	+3V3
10	+5V
11	+3V3
12	+5V
13	+3V3
14	+5V
15	+3V3
16	+5V
17	+3V3
18	+5V
19	+3V3
20	+5V
21	+3V3
22	+5V
23	+3V3
24	+5V

J22 Grounds

Pin	Voltage
1	GND
2	GND
3	GND
4	GND
5	GND
6	GND
7	GND
8	GND
9	GND
10	GND
11	GND
12	GND
13	GND
14	GND
15	GND
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19	GND
20	GND
21	GND
22	GND
23	GND
24	GND

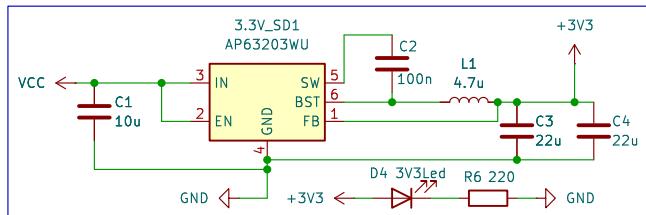
		J25 GPIO	
PC15_D	1	48	PG7_D
PD7_D	2	47	PG8_D
PD10_D	3	46	PG9_D
PD11_D	4	45	PG10_D
PD12_D	5	44	PG11_D
PD13_D	6	43	PG15_D
PD14_D	7	42	ADC123_IN2 PA2_A
PD15_D	8	41	ADC123_IN3 PA3_A
PE3_D	9	40	ADC12_IN4 PA4_A
PE4_D	10	39	ADC12_IN8 PB0_A
PE9_D	11	38	ADC12_IN9 PB1_A
PE10_D	12	37	ADC123_IN10 PC0_A
PE11_D	13	36	ADC123_IN11 PC1_A
PE12_D	14	35	ADC123_IN12 PC2_A
PE13_D	15	34	ADC123_IN13 PC3_A
PE14_D	16	33	ADC12_IN4 PC4_A
PE15_D	17	32	ADC3_IN9 PF3_A
PF2_D	18	31	ADC3_IN14 PF4_A
PF11_D	19	30	ADC3_IN15 PF5_A
PF12_D	20	29	ADC3_IN16 PF6_A
PF13_D	21	28	ADC3_IN8 PF10_A
PF14_D	22	27	×
PF15_D	23	26	×
PG6_D	24	25	×

Debug LED1 D18
MCU_LED1 R39 220 → +3.3V

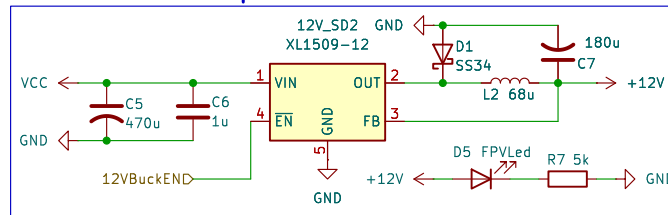
Debug LED2 D19
MCU_LED2 R57 220 → +3.3V

SDMicro LED D20
SD_LED R58 220 → +3.3V

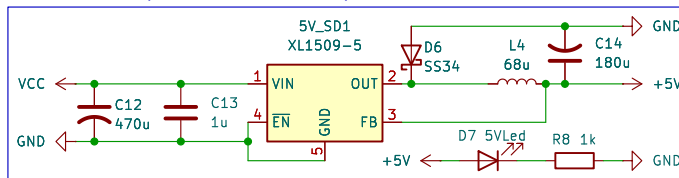
3.3V Logic Step-Down



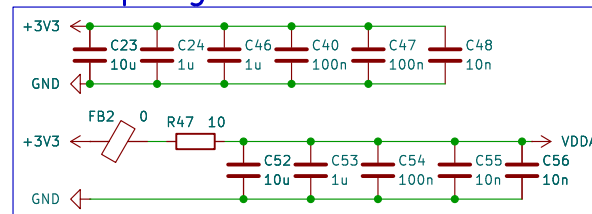
12V FPV Step-Down



5V Component Step-Down



Decoupling



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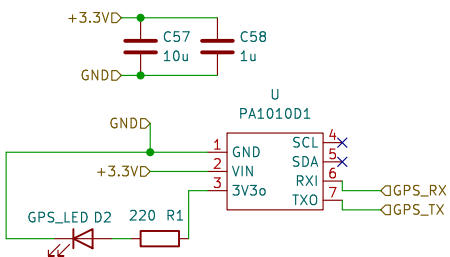
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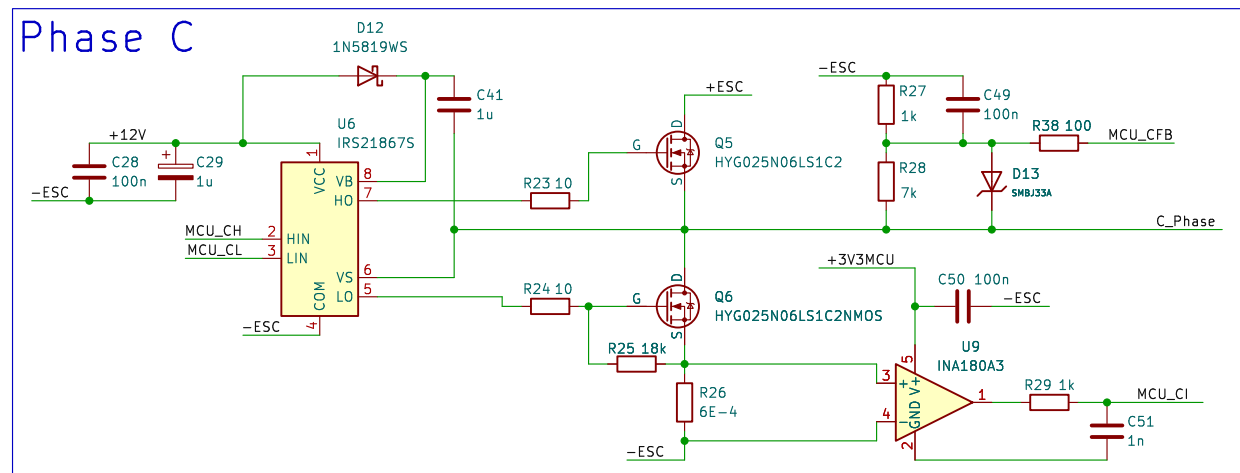
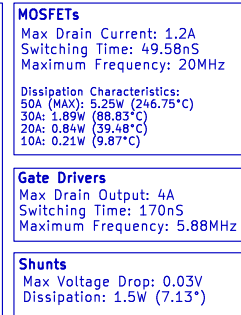
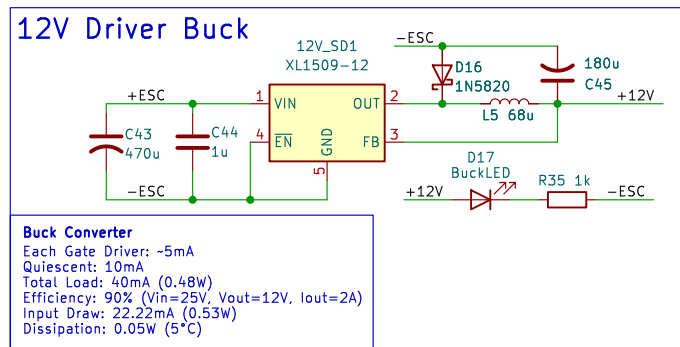
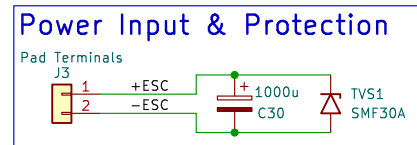
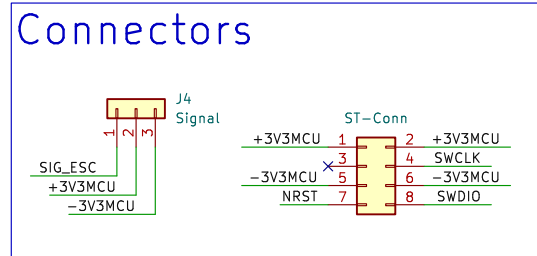
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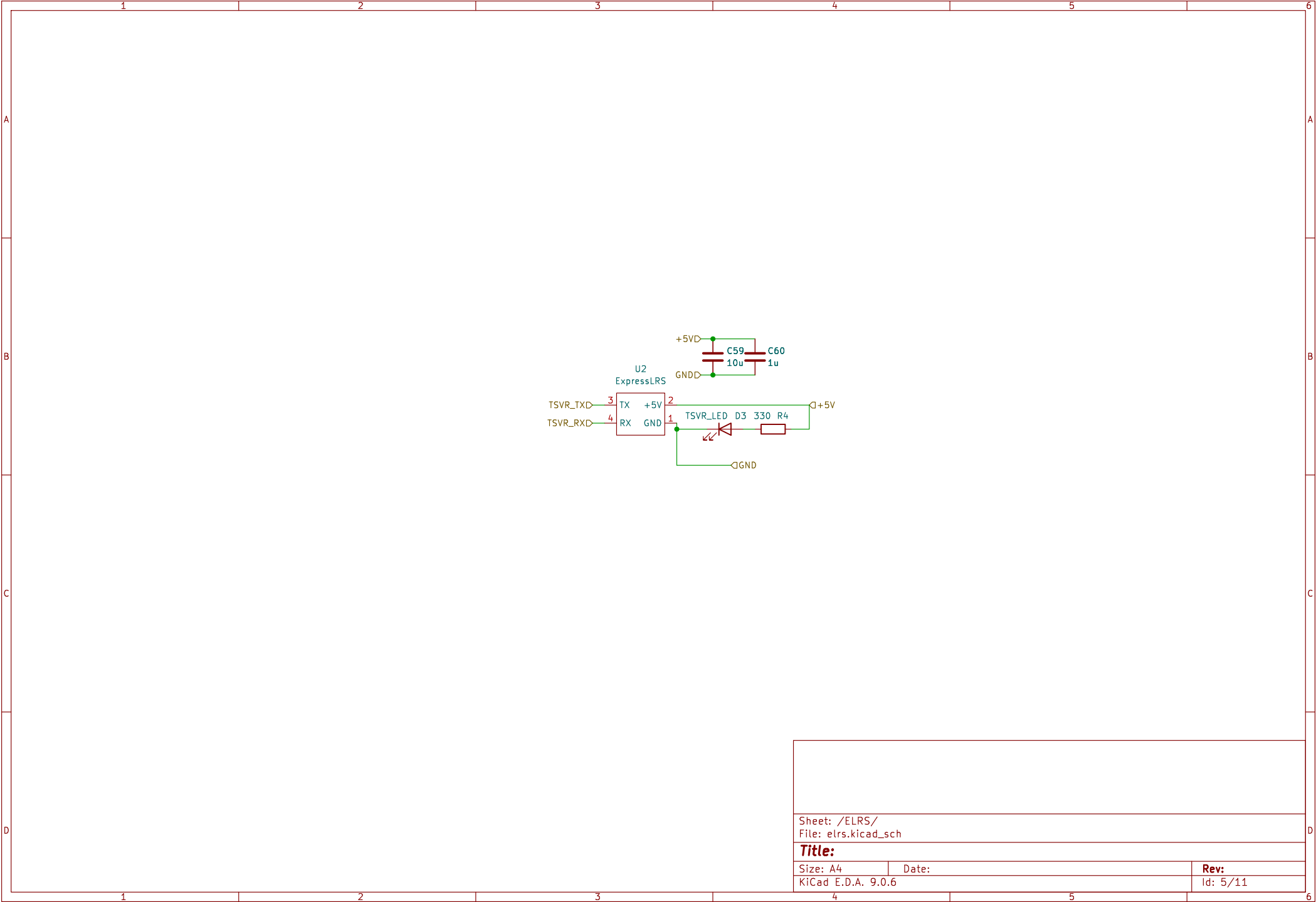
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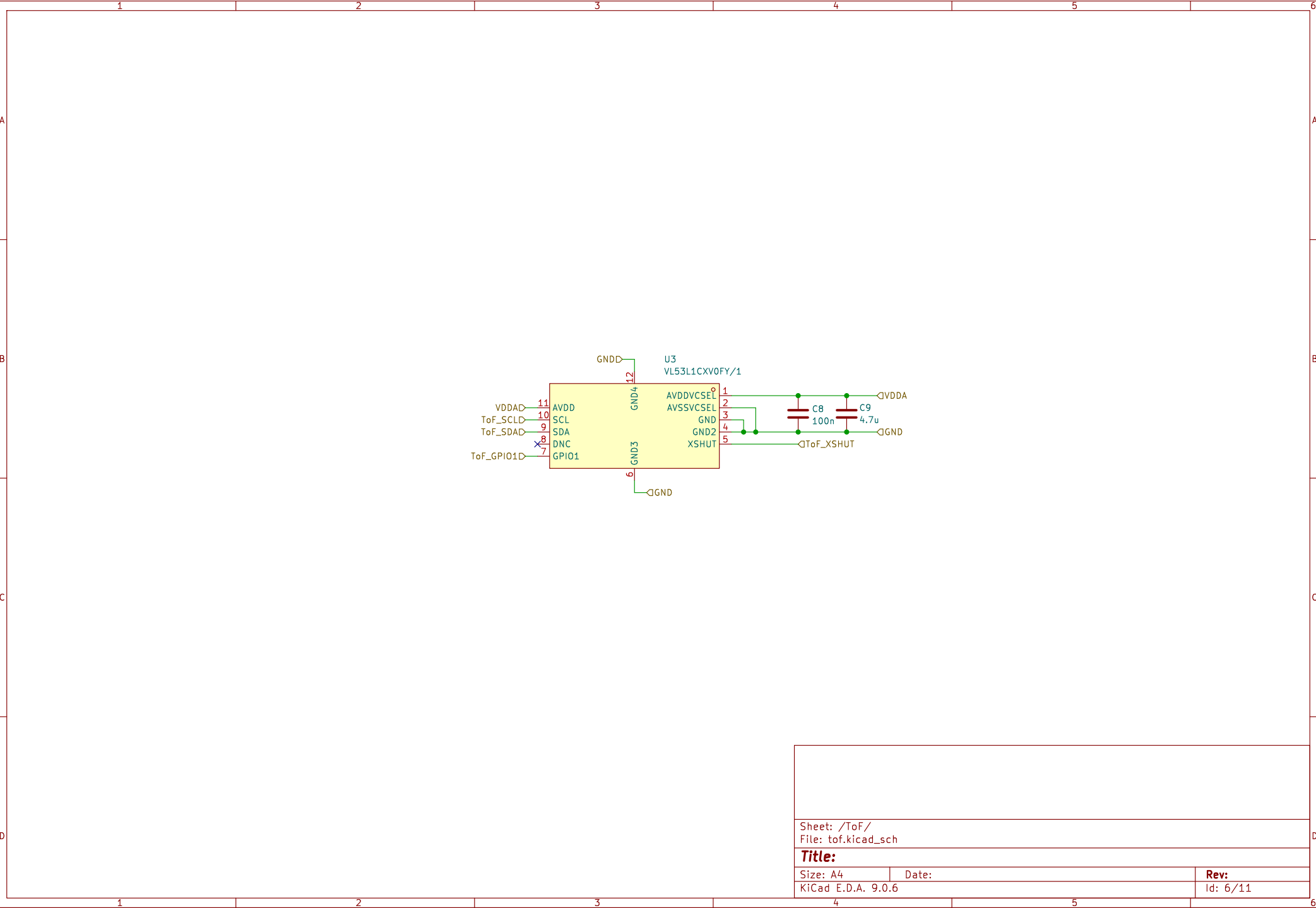
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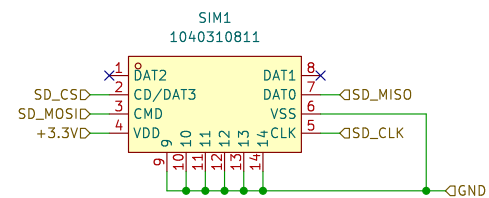
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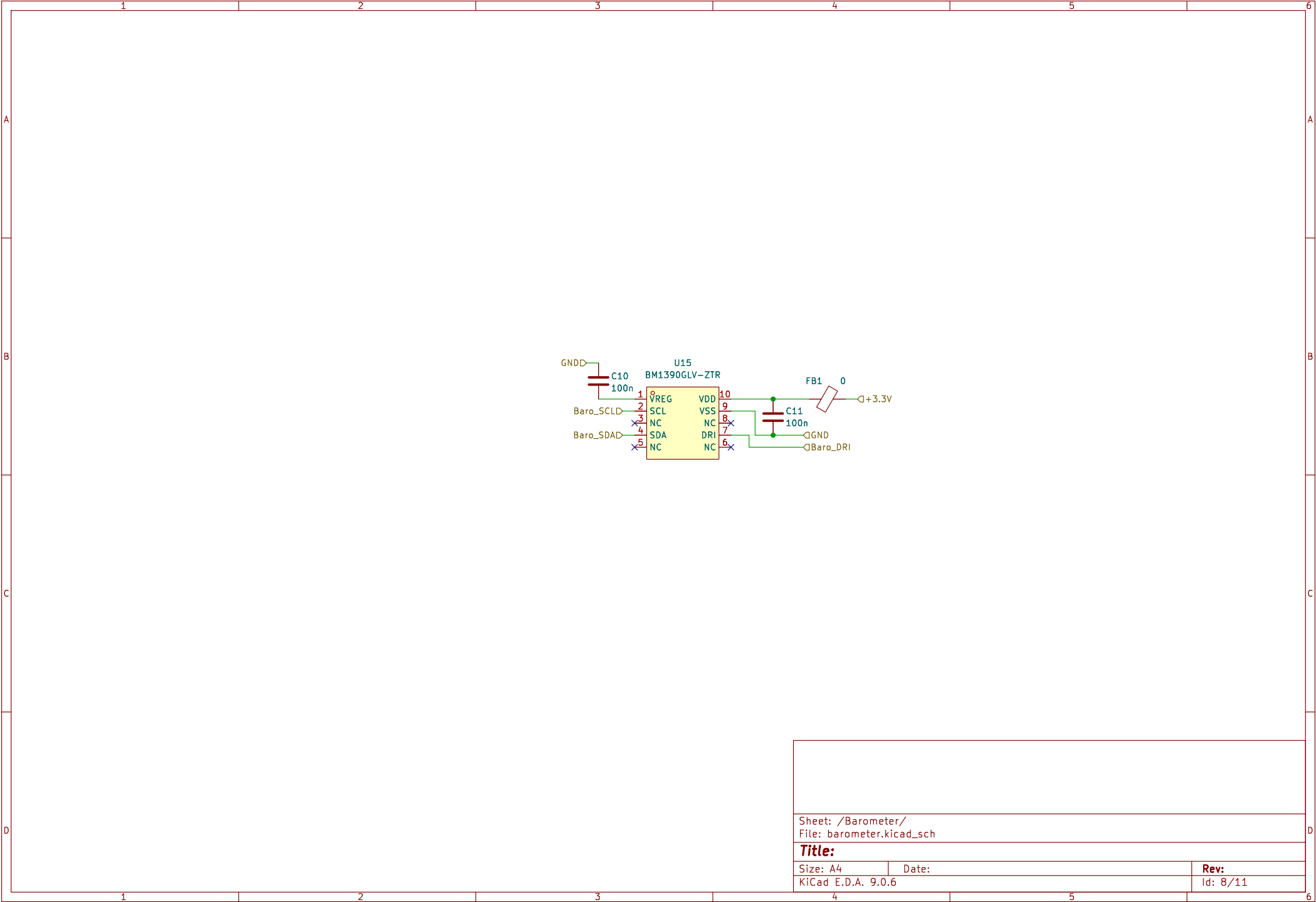
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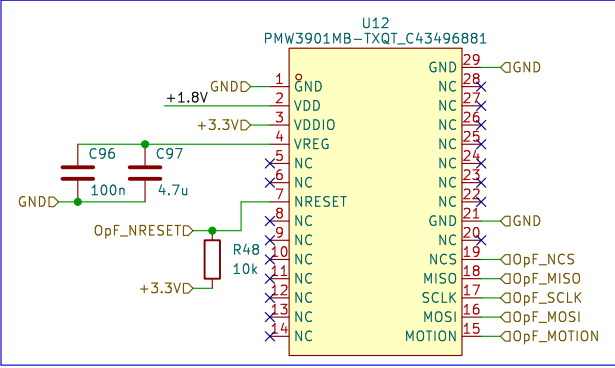
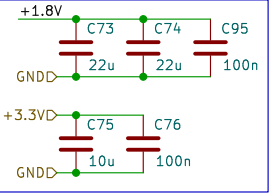
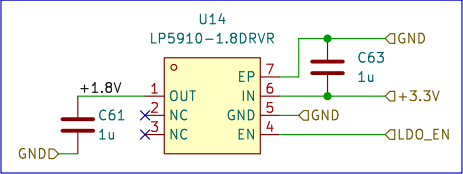
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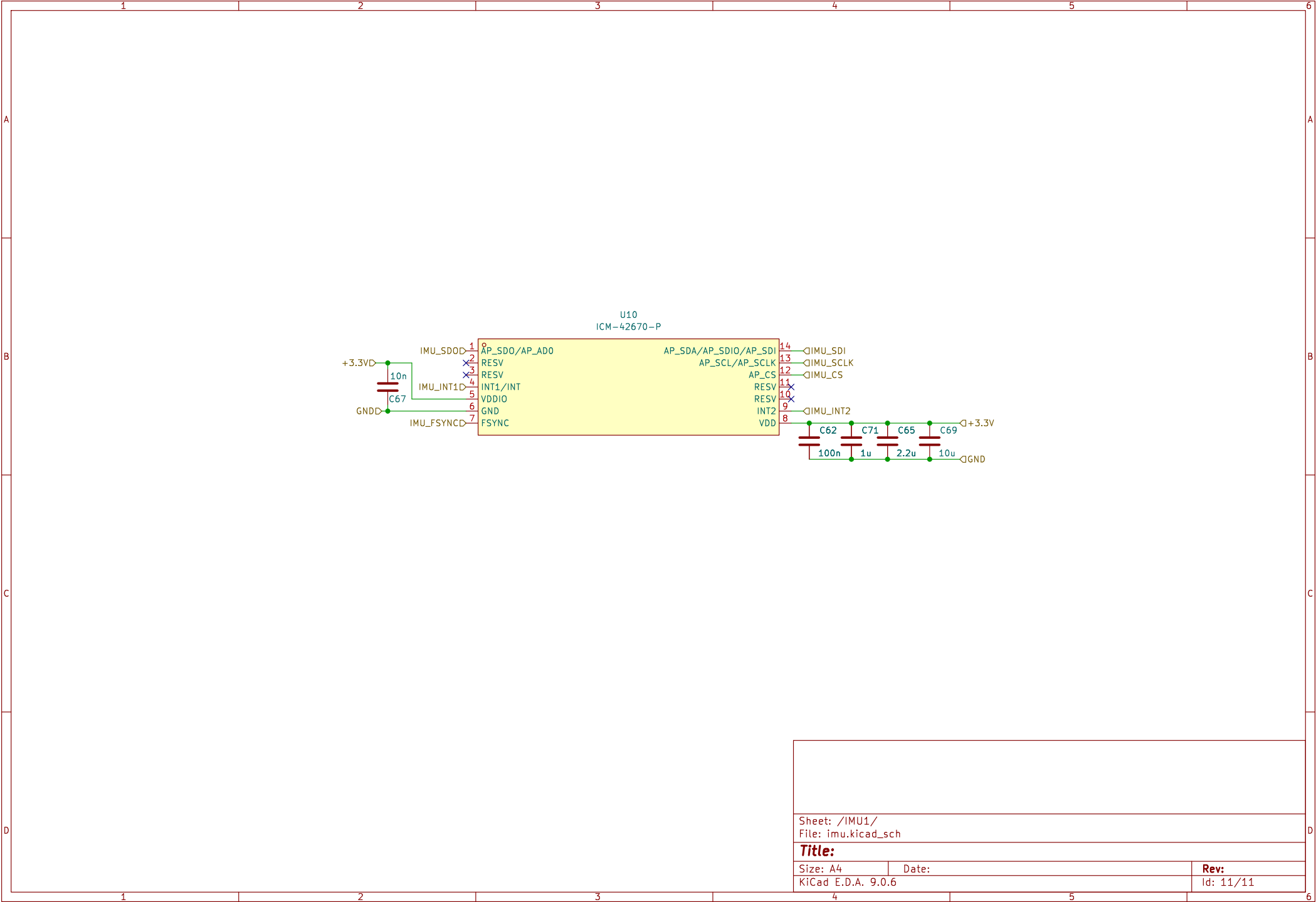
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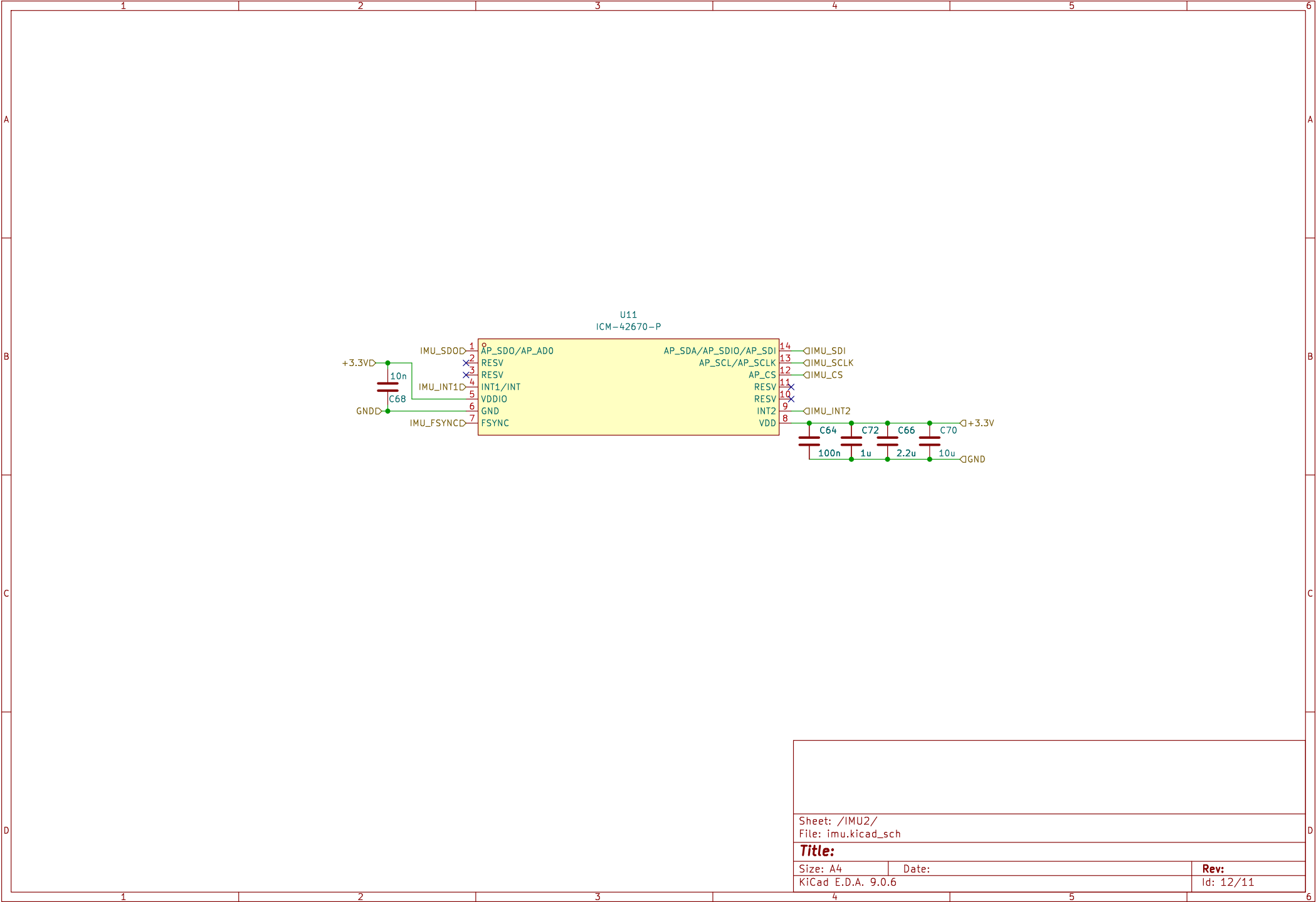
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